INTERNSHIP REPORT

ON

TWITTER ANALYTICS DASHBOARD USING POWER BI

Data Analytics Internship

Submitted to

ElevanceSkills Technology Private Limited

Submitted by

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Role: Data Analytics Intern

Institution: DR. Ambedkar College, Nagpur

Academic Year: 2026–27

Project Type: Data Analytics and Business Intelligence

DECLARATION

I, Aditya Khamankar, declare that this report presents the work completed for the Twitter Analytics Dashboard project during my Data Analytics internship with ElevanceSkills Technology Private Limited.

Name: Aditya Khamankar

Date: 23-10-26

ACKNOWLEDGEMENT

I would like to thank ElevanceSkills Technology Private Limited for the opportunity to work on a practical data analytics project. The project provided experience in preparing data, using Power BI visuals and measures, applying filters, and interpreting social-media engagement metrics. I also acknowledge the guidance and learning resources provided during the internship.

EXECUTIVE SUMMARY

This project uses Power BI to analyze tweet records and present engagement metrics through six dashboard pages. The visuals cover interaction types by tweet category, engagement rate by app-open category, media activity by day of week, replies, retweets and likes, monthly engagement-rate trends, and tweet-level engagement rankings.

The dashboard brings together Power Query data preparation, DAX calculations and Power BI visualizations to summarize tweet interactions and compare performance. The report explains the displayed results and their meaning in the context of the selected fields and filters.

TABLE OF CONTENTS

Section	Page (approx.)
1. Introduction	3
2. Background and Project Purpose	3
3. Learning Objectives	4
4. Tools and Technologies	4
5. Dataset Description	5
6. Project Methodology	5
7. Activities and Tasks	6
8. Skills Developed	12
9. Challenges and Validation Notes	12
10. Feedback and Evidence	13
11. Outcomes and Impact	13
12. Conclusion	14
13. References	14
14. Appendix	14

1. INTRODUCTION

Social-media platforms produce interaction data that can help analysts understand how audiences respond to content. Metrics such as likes, replies, retweets, URL clicks, profile clicks, hashtag clicks, media views and media engagements represent different types of user activity. A dashboard makes these metrics easier to compare and can help identify which tweet categories or individual posts receive more interaction.

This internship project uses Power BI to organize tweet-level data into six visual analyses. The work combines data preparation, filtering, measures and visual presentation. The report follows the dashboard evidence supplied by the project owner and separates direct observations from items that need further validation.

2. BACKGROUND AND PROJECT PURPOSE

The purpose of the project is to convert tweet records into a set of readable visual summaries. Instead of reviewing individual rows manually, the dashboard groups interactions by tweet category, compares engagement rates, shows media activity by weekday, compares replies/retweets/likes, tracks monthly engagement rate and ranks tweets by total engagement.

The analysis is descriptive. It reports the values shown by the available dashboard; it does not by itself prove that a particular tweet feature caused higher engagement. Interpretation should account for the applied filters, the date range, the completeness of the source data and whether the chart is displaying all intended categories.

3. LEARNING OBJECTIVES

• Prepare and organize tweet-level data for analysis.

• Use Power BI fields, filters and measures to summarize engagement metrics.

• Choose visuals appropriate to category comparisons, trends and rankings.

• Interpret the difference between interaction counts and engagement rates.

• Check visual outputs against assignment conditions and investigate unexpected values.

• Document analytical results, limitations and evidence in a structured project report.

4. TOOLS AND TECHNOLOGIES

Tool	Purpose in the project
Microsoft Power BI	Dashboard pages, visualizations, filters and report layout.
Power Query	Data loading and preparation where used in the PBIX workflow.
DAX	Measures and calculated logic for engagement metrics and category comparisons; exact formulas should be checked against the PBIX.
GitHub / project hosting	Project files and supporting documentation, where included in the submission.

5. DATASET DESCRIPTION

The dashboard screenshots indicate tweet-level records with fields or measures associated with tweet category, user profile clicks, hashtag clicks, URL clicks, app opens, engagement rate, media views, media engagements, replies, retweets, likes, tweet text, date/month and total engagement. The exact source column names, record count, data provenance and data types must be confirmed from the dataset or Power BI model before final submission.

6. PROJECT METHODOLOGY

• **Data loading:** Load the Twitter dataset into Power BI and inspect the available fields.

• **Data preparation:** Check data types, missing values, text fields, date fields and numeric engagement columns.

• **Filtering:** Apply the task-specific conditions from the internship assignment. Each task should use its own filter logic where requirements differ.

• **Calculation:** Use measures or calculated columns for the requested engagement metrics and aggregated totals.

• **Visualization:** Select charts according to the question—bar charts for category comparisons, line/area charts for monthly trends, and ranked bars for top tweets.

• **Validation:** Compare visible values with the expected assignment conditions, inspect blank or zero outputs and confirm the active date/time window.

• **Documentation:** Record the result, explain its meaning, capture a screenshot and state limitations.

7. ACTIVITIES AND TASKS

The following task chapters document the six submitted dashboard screenshots. The requirements listed here describe the visible analytical purpose and available evidence. Exact assignment filters and DAX formulas should be cross-checked against the original task sheet and PBIX before final submission.

7.1 TASK 1 — TWEET INTERACTION BREAKDOWN BY CATEGORY

Objective. Compare user-profile clicks, hashtag clicks and URL clicks across tweet categories to understand which interaction types are most prominent in each category.

Visual and implementation. The screenshot uses a horizontal clustered bar chart with tweet category on the category axis and total interactions on the value axis. The three series are Sum of user profile clicks, Sum of hashtag clicks and Sum of URL clicks. The displayed categories are Media Tweet, Other Tweet, Hashtag Tweet and Link Tweet.

Tweet category	Profile clicks	Hashtag clicks	URL clicks
Media Tweet	992	91	525
Other Tweet	458	0	0
Hashtag Tweet	80	107	0
Link Tweet	70	1	718

Result and interpretation. Media Tweets show the largest profile-click total (992) and also have 525 URL clicks and 91 hashtag clicks. Link Tweets have 718 URL clicks, the largest URL-click value among the displayed categories, while Hashtag Tweets have 107 hashtag clicks, the largest hashtag-click value shown. Other Tweets have 458 profile clicks and zero shown for the other two interaction types. This suggests that the dominant interaction type differs by tweet category in the displayed, filtered data.

Validation note. The screenshot reports aggregated totals but does not prove that all assignment filters and the required visual-visibility time windows are active. Verify the even-date and word-count conditions, and capture the relevant filter pane or DAX logic as evidence.

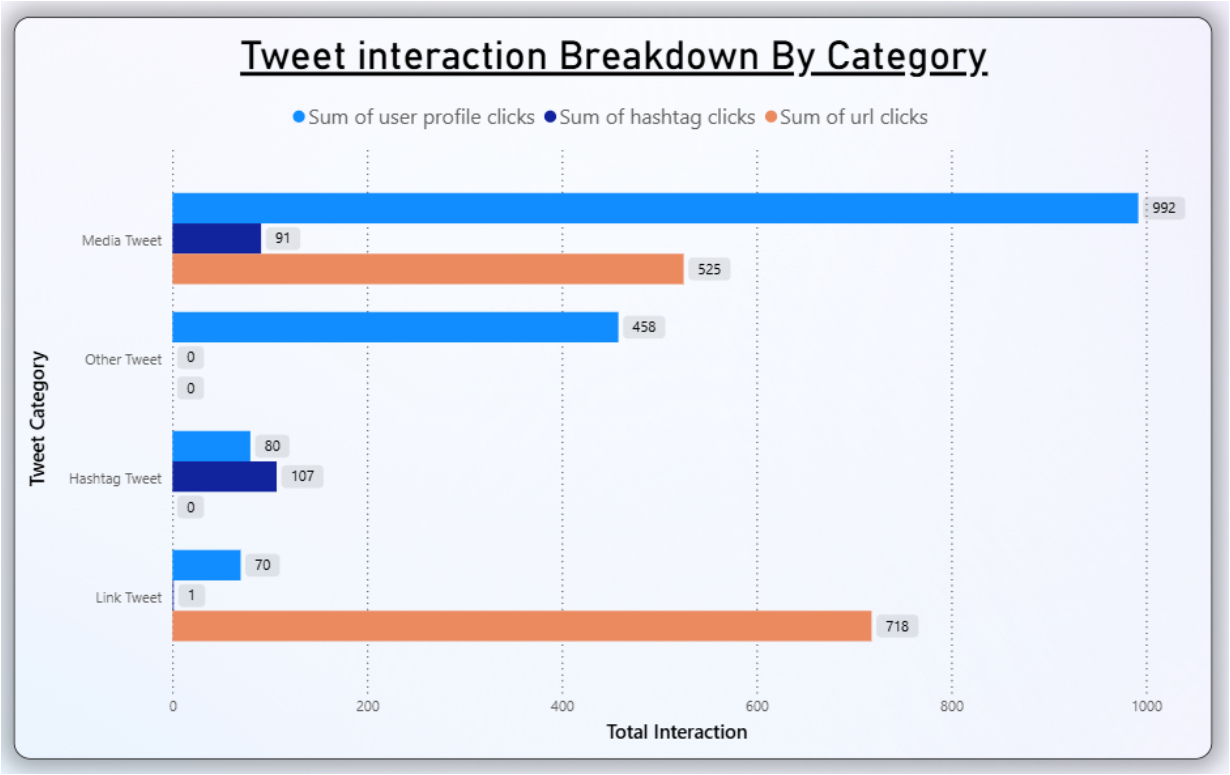


Figure 1. Tweet interaction breakdown by category (user-provided screenshot).

7.2 TASK 2 — ENGAGEMENT RATE: APP OPENS VS NO APP OPENS

Objective. Compare average engagement rate for tweets with app opens and tweets without app opens, using the assignment's specified date, weekday, text and impression filters.

Visual and implementation. The submitted screenshot is a single-column chart. The only displayed category is “Without App Opens,” with a labelled average engagement rate of 0.021 (approximately 2.1% if the underlying measure is a proportion).

Result and interpretation. For the displayed “Without App Opens” group, the average engagement rate is 0.021. However, the visual does not show a value for “With App Opens,” so a comparison cannot yet be made. The current screenshot supports only a one-group result, not a conclusion that one group performs better than the other.

Validation required before submission. Check whether the app-open field contains both categories after filtering, whether the category column/legend is configured correctly, and whether filters have unintentionally removed all rows with app opens. Verify the assignment's weekday and posting-time limits, even impressions, odd dates, character count above 30, exclusion of tweets containing “D,” and the specified visual time windows.

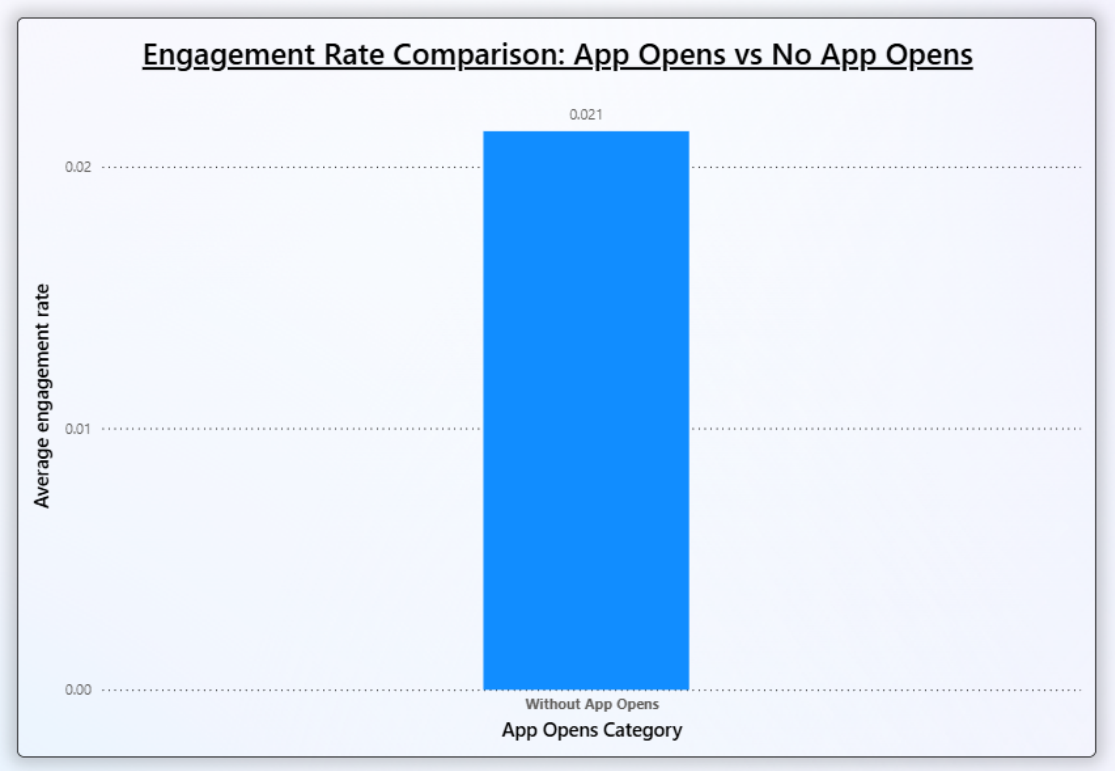


Figure 2. Engagement rate comparison: app opens vs no app opens.

7.3 TASK 3 — MEDIA VIEWS AND ENGAGEMENTS BY DAY OF WEEK

Objective. Compare media views and media engagements by day of week for the requested quarter using a dual-axis visual.

Visual and implementation. The screenshot title says “Media Views and Engagements by Day of Week (Q4).” It displays Monday, Wednesday and Thursday. Media views are shown as 0 on Monday, 0 on Wednesday and 16 on Thursday. The media-engagement line is labelled 0 for all three displayed days.

Day	Media views	Media engagements
Monday	0	0
Wednesday	0	0
Thursday	16	0

Result and interpretation. Thursday is the only displayed day with non-zero media views, at 16. Media engagements are zero on every displayed day. This may reflect the filtered records, but the all-zero engagement series should be checked against the raw media-engagement field and the aggregation settings before drawing a firm conclusion.

Validation note. The screenshot shows only three weekdays, so it does not establish the full weekday pattern. Confirm that the quarter filter corresponds to the requested last quarter, the date and impression filters are correct, tweets containing “H” are excluded as required, and the visual's time-based visibility works in IST.

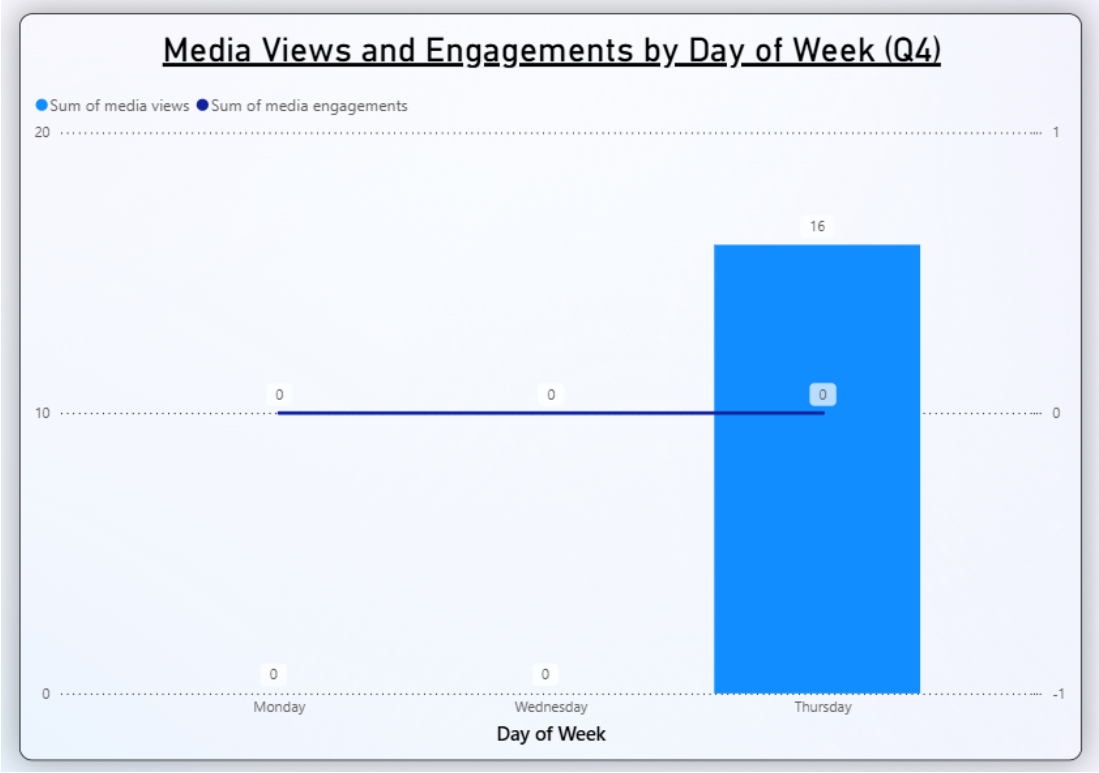


Figure 3. Media views and engagements by day of week (Q4).

7.4 TASK 4 — REPLIES, RETWEETS AND LIKES COMPARISON

Objective. Summarize the three main tweet interaction counts—replies, retweets and likes—to show their relative volumes in the current report context.

Visual and implementation. The submitted chart displays three aggregated measures: Sum of replies, Sum of retweets and Sum of likes. The labels show 666 replies, 375 retweets and 5,747 likes.

Metric	Displayed total
Replies	666
Retweets	375
Likes	5,747

Result and interpretation. Likes are the largest interaction total in this visual: 5,747, compared with 666 replies and 375 retweets. This indicates that, in the displayed records, users interacted through likes much more frequently than through replies or retweets. These are totals, not per-tweet averages or rates, so the result should not be interpreted as the typical response to an individual tweet.

Validation note. The visual's category axis shows no meaningful category label, and the chart title does not specify a date range. Confirm whether this visual is intended to be a standalone total comparison or whether the assignment requires a specific date filter (the PBIX page was noted as having a June–August 2020 filter).

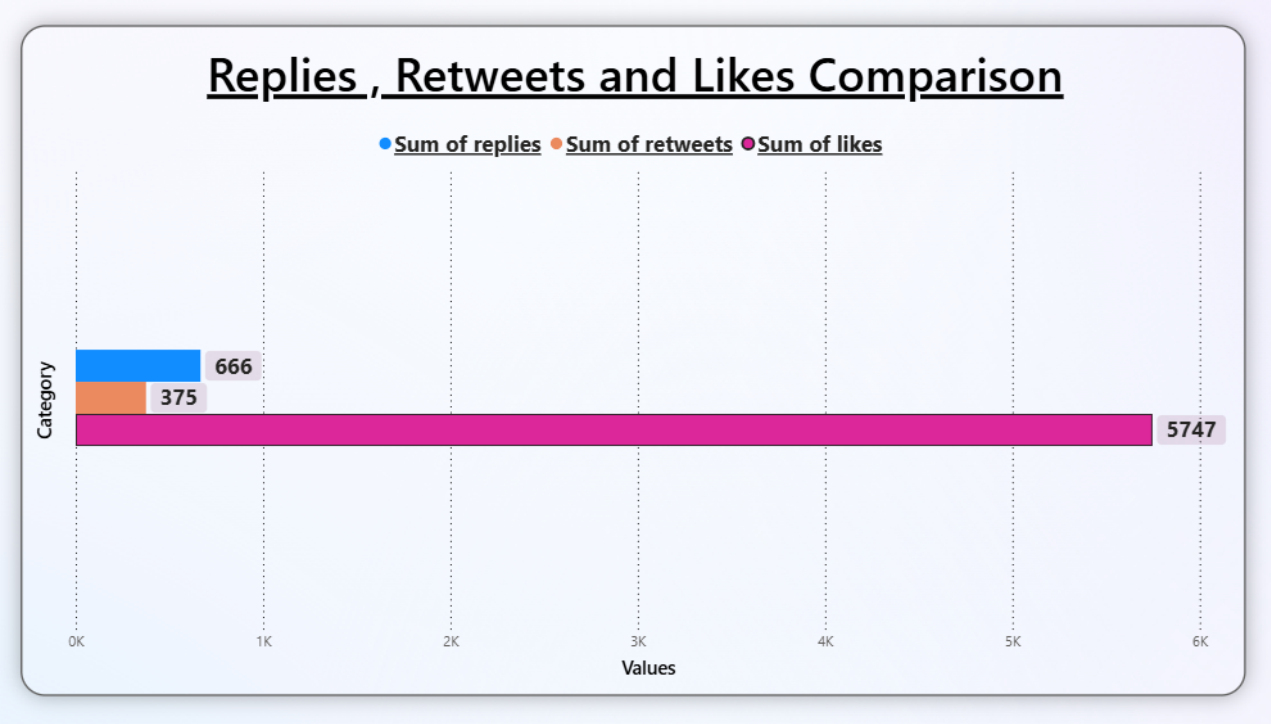


Figure 4. Replies, retweets and likes comparison.

7.5 TASK 5 — MONTHLY AVERAGE ENGAGEMENT RATE TREND

Objective. Compare monthly average engagement rate for tweets with media and without media to identify changes across the year.

Visual and implementation. The screenshot is a monthly line/area-style chart with “No Media” and “With Media” in the legend. The visible labels on the No Media series are approximately 0.8 in January, 0.3 in February, 1.5 in March, 9.0 in May, 0.0 in June, 0.0 in July, 0.1 in August, 0.0 in September and 0.2 in October. April is not displayed, and a visible With Media series cannot be clearly read from this screenshot.

Month	Visible No Media value (approx.)
January	0.8
February	0.3
March	1.5
May	9.0
June	0.0
July	0.0
August	0.1
September	0.0
October	0.2

Result and interpretation. The most prominent visible feature is a spike to approximately 9.0 in May, followed by a sharp decline to 0.0 in June. Because the May value is much larger than the other visible months, it should be verified against the underlying data and engagement-rate formula. Possible causes to check include a small number of records, a scale/formatting issue, an unexpected denominator, or a genuine high-engagement period. The screenshot alone cannot determine which explanation is correct.

Validation note. Confirm that both media categories are present and readable, all required months are included, and the measure is formatted consistently (for example, proportion versus percentage). A monthly trend should be interpreted only after checking the date table, missing months and filter context.

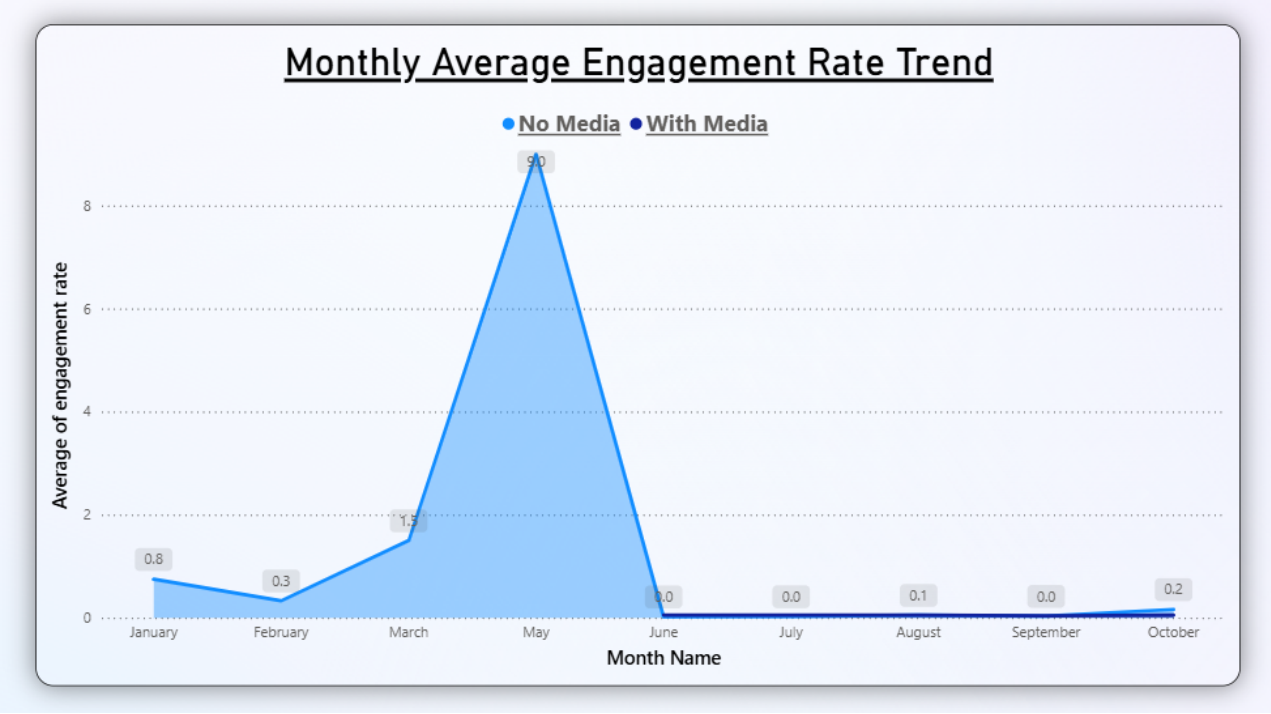


Figure 5. Monthly average engagement rate trend.

7.6 TASK 6 — TOP TWEETS BY TOTAL ENGAGEMENT

Objective. Rank tweets by total engagement so that the posts with the largest combined interaction counts can be identified. The screenshot uses a horizontal bar chart with tweet text as the category and total engagement as the value.

Result. The leading visible tweet begins “justo in hac habitasse platea dictumst et...” and has 88 total engagements. The next visible values are 40, 32, 28, 28, 25, 21, 20, 19, 19 and 19. The screenshot appears to display 11 bars in total, with three bars labelled 19, although the title says “Top 10 Tweets.”

Rank by visible order	Visible tweet text (truncated)	Total engagement
1	justo in hac habitasse platea dictumst et...	88
2	in tempus sit amet sem fusce consequat...	40
3	enim leo rhoncus sed vestibulum sit am...	32
4	faucibus cursus urna ut tellus nulla ut er...	28
5	pellentesque eget nunc donec quis orci ...	28
6	nulla neque libero convallis eget eleifen...	25
7	integer tincidunt ante vel ipsum praesen...	21
8	justo etiam pretium iaculis justo in hac h...	20
9	ut erat curabitur gravida nisi at nibh in h...	19
10	velit nec nisi vulputate nonummy maec...	19
11	vestibulum velit id pretium iaculis diam ...	19

Interpretation. The leading tweet has more than twice the engagement of the second-ranked visible tweet (88 versus 40). There is also a drop from 88 to 40 and then to 32, after which the visible values are clustered between 19 and 28. This suggests that the top tweet stands out within the displayed filtered set. Tweet text is truncated in the chart, so full tweet identification should be taken from a supporting table or tooltip if required.

Validation required. Adjust the visual-level Top N filter or title so the number of displayed records matches the assignment. Also verify the weekday exclusion, associated user/profile field, even-impression and odd-date filters, word-count condition, and specified visibility window.

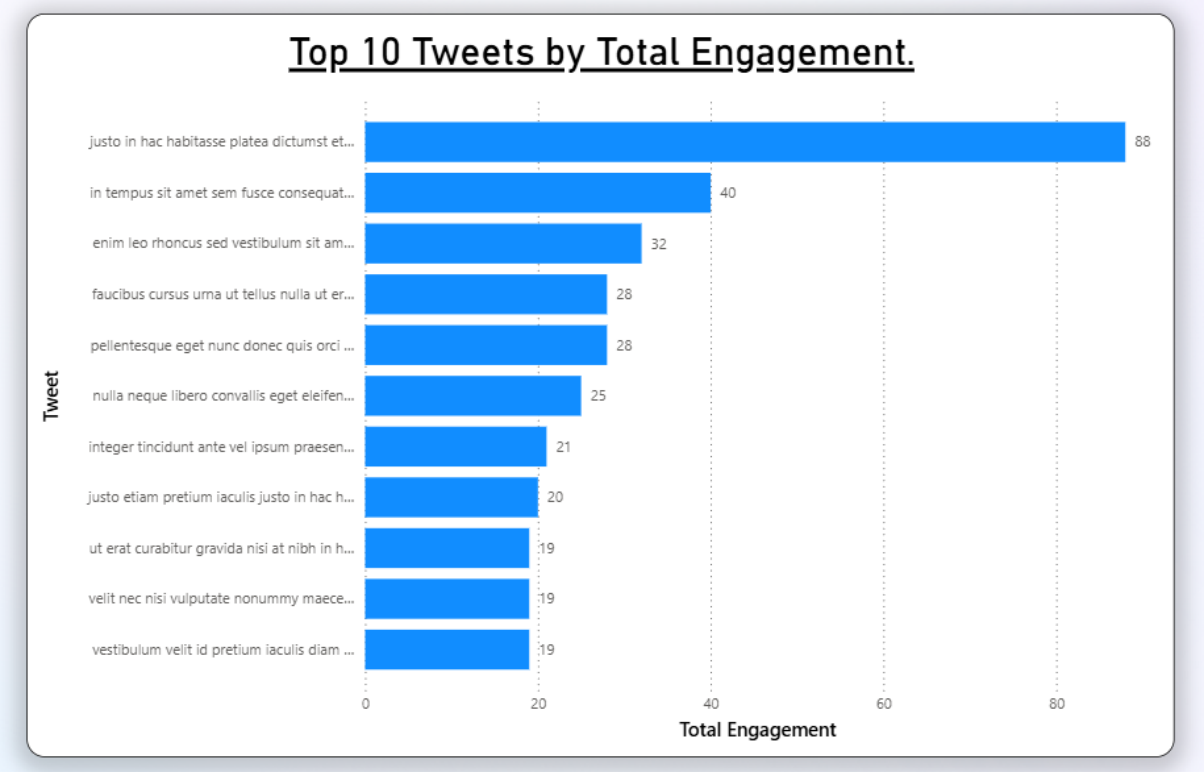


Figure 6. Top tweets by total engagement.

8. SKILLS AND COMPETENCIES DEVELOPED

• **Power BI visualization:** building bar charts, monthly trend visuals and multi-series comparisons.

• **Measure interpretation:** distinguishing aggregate counts from rates and averages.

• **Filtering and segmentation:** analyzing tweet categories and separating records by conditions such as app opens or media presence.

• **Analytical reasoning:** explaining the observed result and identifying where the evidence is insufficient.

• **Validation:** checking missing categories, zero-valued measures, unexpected spikes and Top N configuration.

• **Reporting:** translating dashboard outputs into a structured report with numerical findings and evidence.

9. CHALLENGES AND VALIDATION NOTES

Several points in the current screenshots should be checked before the report is submitted as final. Recording these issues is important because a visual can render successfully while still failing to answer the complete assignment question.

• **Missing comparison group in Task 2:** only “Without App Opens” is visible. Check the data values, filter context, axis/legend setup and measure.

• **Zero media engagements in Task 3:** all visible engagement values are zero. Compare the measure with the source column and check whether the filters leave qualifying records.

• **Monthly spike in Task 5:** the May value is approximately 9.0 while the other visible values are much lower. Confirm the rate formula, unit/format and record count.

• **Top 10 mismatch in Task 6:** the screenshot appears to contain 11 bars. Ensure the visual-level Top N filter returns exactly the required count.

• **Task 4 context:** the screenshot does not clearly identify the category and date scope. Confirm the expected date filter and add a meaningful label or explanatory subtitle.

• **Assignment conditions:** screenshots alone do not prove that every required text/date/impression filter and time-based visibility rule is correctly implemented. Include filter-pane screenshots or describe the exact DAX/filter logic as evidence.

10. FEEDBACK AND EVIDENCE

The evidence currently available for this draft consists of the Power BI dashboard file and six screenshots of the report visuals. The final submission package should include the following items where available:

• Power BI PBIX file.

• Screenshot of each of the six task pages, preferably with page title and relevant filters visible.

• Dataset name/source and a short data dictionary listing the fields used.

• DAX measures and calculated columns relevant to each task.

• GitHub repository and hosted project URL, if part of the submission.

• Internship task assignment/instructions and portal progress evidence, if requested by the evaluator.

• A final verification pass showing that each task meets all specified filter and visibility requirements.

11. OUTCOMES AND IMPACT

The dashboard consolidates multiple engagement measures into a single Power BI project. In the screenshots, likes are the largest of the three interaction totals (5,747), media tweets show high profile-click activity (992), and one tweet leads the total-engagement ranking with 88. These results provide a starting point for exploring differences among interaction types, tweet categories and individual posts.

The review of the screenshots also demonstrates why validation matters. A missing comparison category or an unexpected zero/spike can change the interpretation even when the chart itself looks complete. The report therefore treats these findings as descriptive and identifies the checks needed to strengthen the final submission.

12. CONCLUSION

The Twitter Analytics project provided practical experience in using Power BI to summarize social-media interaction data through category comparisons, engagement-rate visuals, weekday media analysis, interaction totals, monthly trends and tweet rankings. The screenshots provide concrete results that can be reported, but several visuals require verification before their conclusions can be considered complete. Finalizing the report should include checking the original assignment conditions, validating measures and filters, resolving the missing app-open comparison, checking the all-zero media-engagement series, reviewing the monthly spike and ensuring the Top 10 visual contains the intended number of tweets.

13. REFERENCES

• ElevanceSkills internship task assignment and project instructions (add the exact document/title used).

• Twitter dataset supplied for the internship project (add source and access date).

• Microsoft Power BI documentation: https://learn.microsoft.com/power-bi/

• Microsoft DAX documentation: https://learn.microsoft.com/dax/

14. APPENDIX

Appendix A — Project Files

Twitter Dashboard.pbix; six task screenshots supplied for this report; supporting dataset and DAX documentation (to be added or linked in the final submission).

Appendix B — Final Verification Checklist

• Confirm every assignment requirement is mapped to a dashboard element or measure.

• Record the exact DAX formula and aggregation used for each key metric.

• Capture filters and time-window behaviour as evidence.

• Resolve the Task 2 missing category and Task 3 all-zero media-engagement result.

• Validate the Task 5 monthly peak and consistent rate formatting.

• Make the Task 6 Top 10 count exactly match the assignment.

• Update the table of contents and page numbers after final edits.

End of draft report

ADDITIONAL PROJECT EVIDENCE

This appendix supplements the main report with the repository structure, README content and official submission instructions provided by the project owner. The evidence below is based on the uploaded repository screenshot and the text shared in this conversation.

A. GitHub Repository Screenshot

The screenshot shows a repository branch named “main” and the visible files/folders: Images, README.md and Twitter Analytics Dashboard.pbix. This confirms that the screenshot depicts a GitHub-hosted project structure. It does not independently establish that all project files, the dataset, the report or a submission ZIP are present.

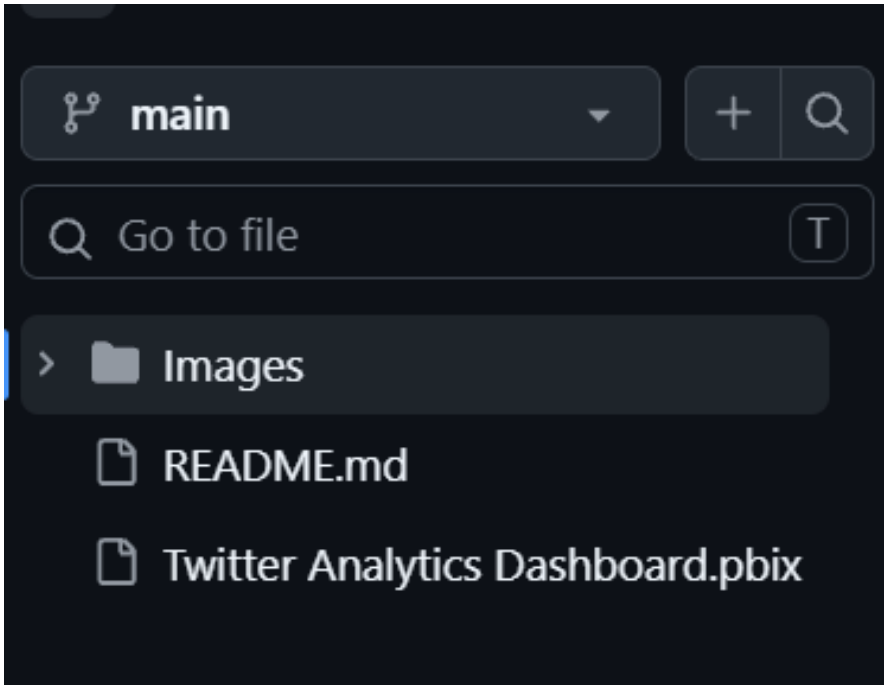


Figure A1. GitHub repository file list supplied by the project owner.

B. README Content — Formatted Screenshot

The following image is a formatted screenshot-style rendering of the README text supplied by the project owner. It summarizes the project overview, objectives, six dashboard features, data preparation, listed DAX fields and author information.

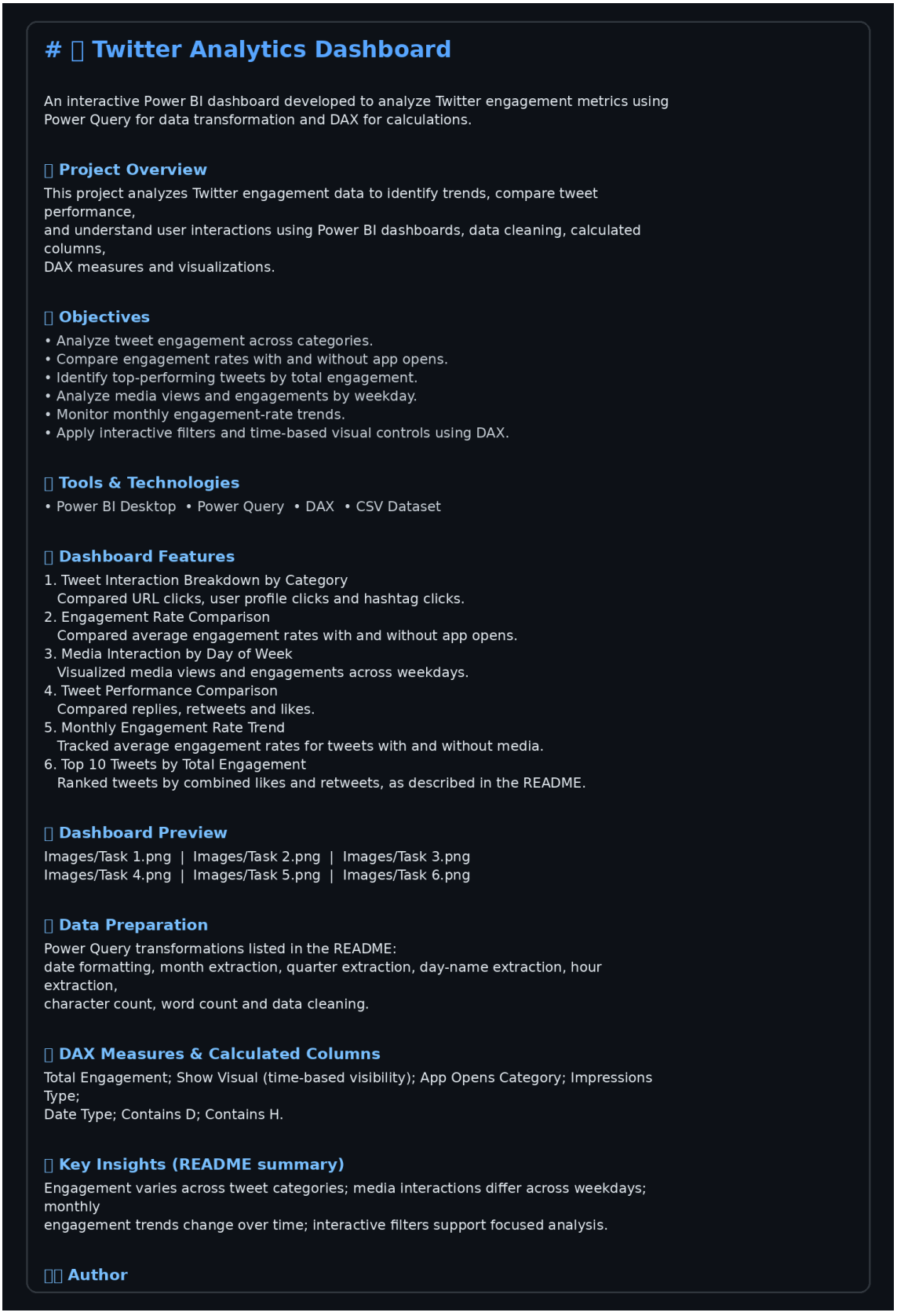


Figure A2. Formatted rendering of the supplied README text; not a direct browser screenshot.